SALMA MBARKI

Software Engineer – Full-Stack, Data & Mobile Developer

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LinkedIn | GitHub | Portfolio

PROFILE

Software engineer (IRM engineering cycle, 2026) specialized in full-stack web development, Flutter mobile applications, real-time data processing and UI/UX design. Experienced through academic internships in software engineering, financial data systems and industrial automation, including two internships at the Tunis Stock Exchange (BVMT). Skilled in building REST-based applications, interactive dashboards and database-driven solutions.

PROFESSIONAL EXPERIENCE

FRTN Technologies, Ariana, Tunisia Aug 2024 – Sep 2024

Robotics & AI Trainer

- Trained children aged 8–16 in robotics and artificial intelligence.
- Designed educational activities and led hands-on robotics and AI workshops.

ACADEMIC INTERNSHIPS

Tunis Stock Exchange (BVMT) Feb 2026 – May 2026

Final Year Project Intern

- Designed and developed an intelligent web platform for real-time stock market data monitoring and visualization.
- Implemented market data processing using OPTIQ/SBE and developed interactive dashboards.
- *Tech: FastAPI, React.js, MongoDB, JWT, Docker, OPTIQ/SBE.*

Tunis Stock Exchange (BVMT) Jul 2025 – Aug 2025

Engineering Intern

- Processed OPTIQ trading data using Python and stored structured information in PostgreSQL.
- Developed an interactive React.js dashboard for financial data visualization.
- *Tech: Python, PostgreSQL, React.js, OPTIQ.*

ArabSoft Jun 2024 – Jul 2024

Full-Stack Developer Intern

- Developed an HR management website for employees, roles, departments and leave requests.
- *Tech: React.js, Spring Boot, MySQL.*

Sagemcom Feb 2023 – May 2023

Final Year Project Intern

- Programmed an automatic screwing machine using Ladder logic and contributed to industrial process optimization.
- *Tech: TIA Portal, WinCC, Ladder.*

Adwya Laboratories Jan 2022 – Feb 2022

Technical Intern

- Detected and resolved production machine failures in collaboration with technical teams.

TECHNICAL SKILLS

- **Programming:** Python, Java, JavaScript, C, C++, HTML, CSS.
- **Full-Stack Development:** React.js, FastAPI, Spring Boot, Node.js, Express.js.
- **Mobile Development:** Flutter, Firebase.
- **Web Services:** REST, GraphQL, SOAP, JWT.
- **Data & Databases:** PostgreSQL, MongoDB, MySQL, NoSQL, Data Visualization.
- **Tools:** Docker, VS Code, IntelliJ, Eclipse, UML.
- **Industrial Automation:** TIA Portal, WinCC, Arduino, Proteus 8.
- **Design:** UI/UX, Interface Design, Interactive Dashboards.

PERSONAL & ACADEMIC PROJECTS

Real-Time Stock Market Platform – 2026

Developed an intelligent web platform to process, monitor and visualize real-time financial market data.

Tech: FastAPI, React.js, MongoDB, JWT, Docker, OPTIQ/SBE.

Web Services Project

Developed a website integrating three web service architectures: REST for academic documents management, GraphQL for visa management and SOAP for medical records management.

Tech: REST, GraphQL, SOAP.

E-Commerce Mobile Application – 2025
Built a mobile shopping application providing a modern e-commerce experience.
Tech: Flutter, Firebase.

Insurance Management Website – 2024
Developed a full-stack web application for insurance management.
Tech: React.js, Express.js, MongoDB.

Human Resources Management Website – 2024
Built a web platform for managing employees, roles, departments and leave requests.
Tech: React.js, Spring Boot, MySQL.

Automatic Screwing Machine – 2023
Programmed and optimized an automated industrial screwing machine.
Tech: TIA Portal, WinCC, Ladder.

EDUCATION

International Private Polytechnic School of Tunis 2023 – 2026
Engineering Cycle – Computer Science, Networks & Multimedia (IRM)

ISET Rades, Tunisia 2020 – 2023
Applied Bachelor's Degree – Electrical Engineering, Automation & Industrial Computing

Carthage Presidency High School 2019 – 2020
High School Diploma – Technical Sciences

CERTIFICATIONS

- ITS Python – 2025.
- DELF B2 – 2025.
- ITS Java – 2024.
- ITS JavaScript – 2024.
- ITS Databases – 2024.
- Microsoft Office Specialist & Associate – 2024.

LANGUAGES

Arabic (Native) | French (DELF B2) | English (Conversational)